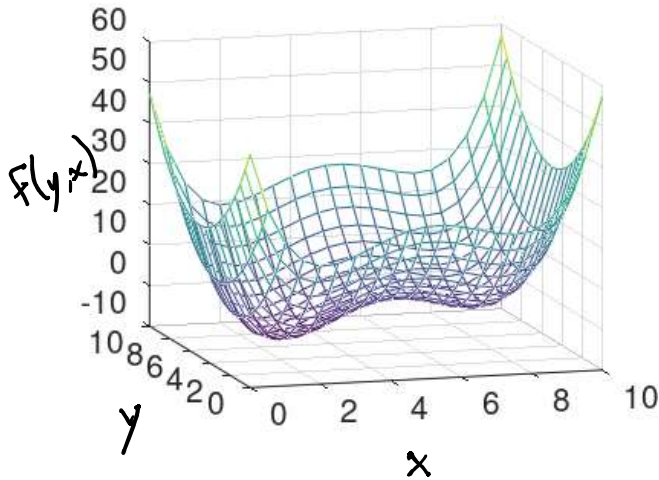




Suppose we have a function $f(x_1, \dots, x_n)$
Taylor expansion...

$$f(\vec{x} + \vec{\epsilon}) = f(\vec{x}) + \vec{\epsilon}^T \nabla f(\vec{x})$$

To minimize $f(\vec{x} + \vec{\epsilon})$ we minimize
 $\vec{\epsilon}^T \nabla f(\vec{x})$

Assume $\vec{\epsilon}$ is of fixed magnitude

$$\vec{\epsilon}^T \nabla f(\vec{x}) = |\vec{\epsilon}| |\nabla f(\vec{x})| \cos \Theta$$

This is minimized when $\cos \Theta = -1$
 $\Theta = 180^\circ$

$\vec{\epsilon}$ is opposite $\nabla f(\vec{x})$

Gradient Descent (multi-variable)

Given a function $f(\vec{x})$, its minimum can be approximated by:

1. Guess initial value $\vec{x}$, step size α .

2. Compute $\nabla f(\vec{x})$.

3. Update $\vec{x} \leftarrow \vec{x} - \alpha \nabla f(\vec{x})$

(update $x_i \leftarrow x_i - \alpha \frac{\partial}{\partial x_i} f(\vec{x})$ for all i)

4. Repeat until convergence (change in $f(\vec{x})$ is less than threshold).

Example

$$f(x, y) = x^2 y^2 + xy + 1$$

Initial: $[1, -1]$

Step size: $\alpha = 0.1$

$$\nabla f(\vec{x}) = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

$$\nabla f(\vec{x}) = [2xy^2 + y, 2yx^2 + x]$$

Initially:

$$\vec{x} = [1, -1]$$

$$f(\vec{x}) = 1$$

$$\nabla f(\vec{x}) = [1, -1]$$

Iteration 1:

$$\vec{x} \leftarrow [1, -1] - 0.1 [1, -1]$$

$$\vec{x} \leftarrow [0.9, -0.9]$$

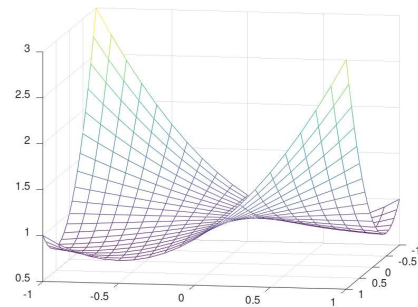
$$f(\vec{x}) = 0.846$$

$$\nabla f(\vec{x}) = [0.558, -0.558]$$

Iteration 2:

$$\vec{x} \leftarrow [0.9, -0.9] - 0.1 [0.558, -0.558]$$

$$\vec{x} \leftarrow [0.844, -0.844]$$



$$f(\vec{x}) = 0.795$$

$$\nabla F(\vec{x}) = [0.359, -0.359]$$

Iteration 3:

$$\vec{x} \leftarrow [0.844, -0.844] - 0.1[0.359, -0.359]$$

$$\vec{x} \leftarrow [0.808, -0.808]$$

$$f(\vec{x}) = 0.774$$

⋮

$$\text{True minimum: } \vec{x} = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$$

$$f(\vec{x}) = 0.75$$

Challenges:

1. Overstepping minimum.
2. Converging to local minimum.
3. How to choose threshold?

We want to find a set of weights $\vec{w}$ that yields outputs as close as possible to the targets.

that minimize loss function for a set of input-target pairs.

We want to minimize L , with respect to parameters $\vec{w}$.

Gradient Descent (neural networks)

Given a loss function L , the optimal set of weights $\vec{w}$ can be found by:

Given a loss function L , the optimal set of weights $\vec{w}$ can be found by:

1. Choose an initial $\vec{w}$, step size α .
2. Compute $\nabla L(\vec{w})$ on a set of instances from the training set.
3. Update $\vec{w} \leftarrow \vec{w} - \alpha \nabla L(\vec{w})$.
(update $w_i \leftarrow w_i - \alpha \frac{\partial}{\partial w_i} L(\vec{w})$ for all i)
4. Repeat until change in $L(\vec{w})$ is less than a threshold.

Problem 1. Overshooting.

Learning Rate Schedule - Definition of how the learning rate α changes as a function of the number of iterations.

Factor Schedule:

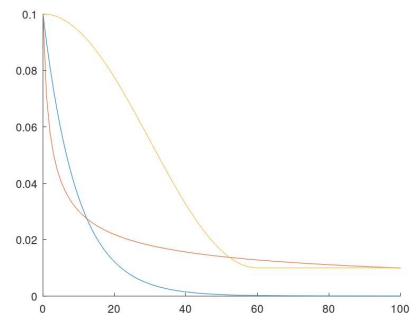
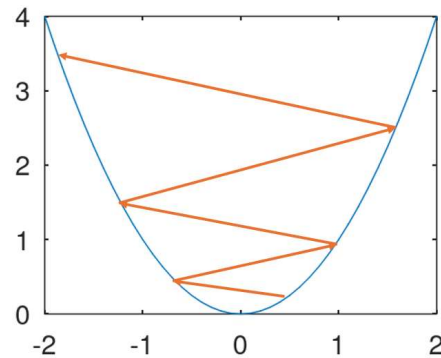
$$\alpha_i = \alpha_{i-1} \cdot \beta \quad \text{where } \beta \in (0, 1)$$

Root Schedule:

$$\alpha_i = \frac{\alpha_0}{\sqrt{i+1}}$$

Cosine Schedule:

$$\alpha_i = \begin{cases} \alpha_T + \frac{\alpha_0 - \alpha_T}{2} \left(1 + \cos\left(\frac{\pi i}{T}\right)\right) & \text{if } i < T \\ \alpha_T & \text{if } i \geq T \end{cases}$$



Problem 2. Local Minima.

Stochastic Gradient Descent

stochastic descent, select

Stochastic Gradient Descent

At each iteration in gradient descent, select a random subset of input-targets

This adds randomness $\nabla L(\vec{w})$

Stochastic Gradient Descent:

Subset size is 1.

Mini-Batch Gradient Descent.

Proper subsets of fixed size > 1 .

Momentum

Maintain vector $\vec{v}$ that is a weighted combination of gradients at prior iterations.

Gradient Descent (with momentum)

1. Choose initial $\vec{w}$, learning rate α , momentum constant β .

2. Compute $\nabla L(\vec{w})$.

3. $\vec{v} \leftarrow \beta \vec{v} + (1 - \beta) \nabla L(\vec{w})$

4. $\vec{w} \leftarrow \vec{w} - \alpha \vec{v}$.

5. Repeat until convergence.

